PREBOARDING BATCH - PROJECT CASE STUDY

TELECOM SERVICE ASSURANCE & TROUBLE TICKET MANAGEMENT SYSTEM

Telecom OSS – Incident, Fault, SLA & Support Management

1. Business Scenario

A telecommunications operator manages thousands of mobile sites, network elements, broadband connections and enterprise services.

Customers may experience:

- Network outages
- Slow internet
- Call drops
- SIM-related issues
- Broadband failures
- Service degradation

The company requires a **Telecom Service Assurance & Trouble Ticket Management System (TSATMS)**.

The system should capture incidents, assign them to support engineers, track SLA, escalate critical incidents and maintain complete resolution history.

Roles:

- Customer
- Service Desk Administrator
- Network Engineer
- Network Manager

2. Login

```
==========================================================
   TELECOM SERVICE ASSURANCE SYSTEM
==========================================================
```

1. Customer Login

2. Service Desk Login

3. Network Engineer Login

4. Network Manager Login

5. Forgot Password

6. Exit

Implement:

- CAPTCHA
- Password authentication
- OTP
- Account locking
- Login history
- Role-based access

3. Customer & Service Information

Customer

customerId

customerNumber

customerName

email

mobileNumber

customerType

city

status

Customer types:

CONSUMER

SME

ENTERPRISE

TelecomService

serviceId

serviceCode

serviceName

serviceType

customerId

activationDate

serviceStatus

Services:

MOBILE

BROADBAND

ENTERPRISE_CONNECTIVITY

VPN

CLOUD_CONNECTIVITY

4. Trouble Ticket

A customer or service desk agent can raise a ticket.

TroubleTicket

ticketId

ticketNumber

customerId

serviceId

category

description

priority

severity

createdDate

assignedEngineerId

status

slaDeadline

resolutionDate

Priority:

LOW

MEDIUM

HIGH

CRITICAL

Status:

OPEN

ASSIGNED

IN_PROGRESS

PENDING_CUSTOMER

ESCALATED

RESOLVED

CLOSED

CANCELLED

5. Sample Ticket

Ticket Number : TT-2026-004521

Customer : CUST100245

Service : Enterprise Connectivity

Category : NETWORK_OUTAGE

Priority : CRITICAL

Status : IN_PROGRESS

Engineer : ENG1008

SLA Deadline : 08-Aug-2026 18:30

6. Incident Categories

NETWORK_OUTAGE

CALL_DROP

SLOW_DATA

NO_CONNECTIVITY

SIM_ISSUE

BILLING

BROADBAND

ROAMING

ENTERPRISE_LINK

OTHER

7. Engineer Management

NetworkEngineer

engineerId

employeeCode

engineerName

specialization

region

experienceYears

availability

activeTicketCount

Example:

Engineer	Skill	Region	Experience	Active Tickets
ENG1008	Core Network	West	9 yrs	2
ENG1015	RAN	North	7 yrs	4
ENG1021	Broadband	South	6 yrs	1

ENG1030	IP Network	West	10 yrs	3

The system should recommend an engineer based on:

- Specialization
- Region
- Availability
- Experience
- Current workload

Use:

Stream + Lambda + Comparator + Optional

8. SLA Management

Different priorities have different SLA limits.

Example:

Priority	Response SLA	Resolution SLA
CRITICAL	15 min	2 hours
HIGH	30 min	4 hours
MEDIUM	2 hours	12 hours
LOW	8 hours	48 hours

The system should automatically calculate:

SLA Deadline

SLA Status

Remaining Time

SLA status:

WITHIN_SLA

AT_RISK

BREACHED

9. Ticket Escalation

If a ticket is approaching or exceeding SLA:

Engineer

↓

Team Lead

↓

Network Manager

↓

Operations Manager

EscalationHistory

escalationId

ticketId

fromLevel

toLevel

reason

escalationDate

escalatedBy

Critical tickets should be processed before lower-priority tickets.

Use:

PriorityQueue

10. Incident Resolution

Engineer should be able to update:

Root Cause

Resolution

Resolution Date

Resolution Code

Resolution codes:

HARDWARE_FAILURE

CONFIGURATION_ERROR

NETWORK_CONGESTION

SOFTWARE_FAILURE

FIBER_CUT

POWER_FAILURE

CUSTOMER_DEVICE

UNKNOWN

The system should maintain complete ticket history.

TicketStatusHistory

historyId

ticketId

oldStatus

newStatus

changedBy

changedDate

remarks

11. Network Event Processing

The system should simulate incoming network events.

Example:

Event ID : NE-884521

Network Node : MUM-RAN-045

Event Type : LINK_DOWN

Severity : CRITICAL

Event Time : 08-Aug-2026 16:45

A background thread should process network events and potentially create trouble tickets automatically.

This demonstrates:

- Multithreading
- Queue
- Producer/Consumer concept
- Exception handling
- Synchronization

12. Notifications

Generate notifications for:

- Ticket creation
- Engineer assignment
- SLA warning
- SLA breach
- Ticket resolution
- Ticket closure

Notification

notificationId

recipientId

message

notificationType

createdDate

readStatus

13. Customer Dashboard

1. View Active Services
2. Raise Trouble Ticket
3. View My Tickets
4. Track Ticket

5. View Ticket History

6. View Notifications

7. Submit Feedback

8. Logout

Customer can view:

Ticket Number

Service

Priority

Status

Engineer

SLA Status

Resolution

14. Service Desk Dashboard

1. View Open Tickets

2. Assign Engineer

3. Reassign Ticket

4. Escalate Ticket

5. Update Priority

6. Monitor SLA

7. Close Ticket

8. Generate Reports

15. Network Manager Dashboard

Display:

Total Open Tickets : 248

Critical Incidents : 12

SLA At Risk : 21

SLA Breached : 7

Resolved Today : 184

Average Resolution Time : 3.8 Hours

16. Java 8 Requirements

Use Stream API for:

- Tickets by status
- Tickets by priority
- Engineer workload
- SLA breach analysis
- Average resolution time
- Tickets by region
- Top incident categories
- Engineer performance
- Customers with repeated incidents

Example:

Find the three engineers with the lowest active workload who have the required specialization and are currently available.

17. Multithreading

Implement:

NetworkEventProcessor

Processes network events.

SLA Monitor

Checks tickets approaching SLA deadline.

NotificationProcessor

Processes pending notifications.

ReportGenerator

Generates reports asynchronously.

Use:

ExecutorService

ScheduledExecutorService

BlockingQueue

Runnable

Callable

Future

Synchronization

18. Reports

Generate:

- Ticket Volume Report
- SLA Compliance Report
- Engineer Performance Report
- Incident Category Report
- Regional Incident Report
- Average Resolution Report
- Critical Incident Report

Export to CSV/TXT.

19. JDBC Transaction

Ticket assignment should be transactional:

Validate Ticket

↓

Validate Engineer

↓

Check Engineer Availability

↓

Assign Engineer

↓

Update Ticket

↓

Create Status History

↓

Create Notification

↓

Create Audit Record

↓

COMMIT

Any failure → ROLLBACK.

20. Suggested Classes

Customer

TelecomService

TroubleTicket

NetworkEngineer

SLAConfiguration

TicketStatusHistory

EscalationHistory

NetworkEvent

Notification

Feedback

AuditLog

LoginHistory

AuthenticationService

TicketService

EngineerAssignmentService

SLAService

EscalationService

NetworkEventService

NotificationService

ReportService

CustomerDAO

ServiceDAO

TicketDAO

EngineerDAO

SLAConfigDAO

NetworkEventDAO

NotificationDAO

CaptchaGenerator

OTPService

PasswordUtil

DBConnection

ReportGenerator

COMMON DATABASE REQUIREMENT

Each project must have a properly normalized relational database.

Participants should demonstrate:

Tables

- Primary Keys

- Foreign Keys
- Unique Constraints
- NOT NULL
- Appropriate indexes
- Status columns
- Created/Updated timestamps

SQL

At minimum:

SELECT

INSERT

UPDATE

DELETE

JOIN

LEFT JOIN

GROUP BY

HAVING

ORDER BY

SUBQUERY

CASE

AGGREGATE FUNCTIONS

DATE FUNCTIONS

Where appropriate, participants may implement:

Views

Stored Procedures

Functions

Triggers

Indexes

COMMON JAVA ARCHITECTURE

A suitable package structure is:

```
com.amdocs.telecom
│
├── controller
│
├── service
│   └── impl
│
├── dao
│   └── impl
│
├── model
│
├── dto
│
├── exception
│
├── validation
│
├── security
│
├── scheduler
│
├── report
│
├── util
```

```
│
└── main
```

Participants should demonstrate:

Core Java

- Classes and Objects
- Encapsulation
- Inheritance
- Abstraction
- Polymorphism
- Interfaces
- Enums
- Generics
- Collections

Java 8

- Lambda
- Functional Interfaces
- Stream API
- Optional
- Method References
- Default/Static Interface Methods

JDBC

- Connection
- PreparedStatement
- ResultSet
- Batch Processing
- Transactions
- Commit

- Rollback
- Savepoint

Multithreading

- Thread/Runnable
- ExecutorService
- Callable/Future
- ScheduledExecutorService
- Synchronization
- BlockingQueue

Security

- CAPTCHA
- Password hashing
- OTP
- Account locking
- Role-based authorization
- PreparedStatement

File Handling

- CSV export
- Text reports
- Configuration files
- Logging

Design Patterns

Minimum three, such as:

DAO

Factory

Strategy

Singleton

Observer